



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING

SEMESTER 1 2025/2026

SECP3623 Database Programming

SECTION 02

Assignment 2

LECTURER: DR. SHAMINI A/P RAJA KUMARAN

Student Name	Matric No.
LAM YOKE YU	A23CS0233
TEH RU QIAN	A23CS0191



No. Kad Pengenalan/No. ISID:.....
I.C. No./ISID No.

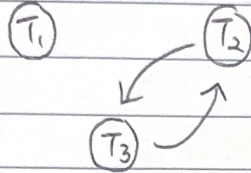
No. Soalan.....
Question No.

Muka surat.....
Page

Jangan tulis apa-apa di ruangan ini
Do not write anything in this column

Question 1

(i)



(ii) Since there is a cycle, the schedule is non conflict-serializable.

(iii)	Time	T_1	T_2	T_3
	t_1	begin transaction		
	t_2	lock-S(X)		
	t_3	read-item(X)		
	t_4	unlock(X)	begin transaction	
	t_5	lock-S(Y)	lock-X(M)	
	t_6	read-item(Y)	read-item(M)	begin transaction
	t_7	unlock(Y)	lock-S(N)	request X-lock(N)
	t_8	commit	read-item(N)	WAIT
	t_9		$M = (N+M) * 10$	WAIT
	t_{10}		write-item(M)	WAIT
	t_{11}		commit/unlock(M)(N)	WAIT
	t_{12}			lock-X(N)
	t_{13}			read-item(N)
	t_{14}			lock-S(M)
	t_{15}			read-item(M)
	t_{16}			$N = N + 150$
	t_{17}			write-item(N)
	t_{18}			commit/unlock(N,M)

(iv) At t_7 , as T_2 and T_3 request for the lock at the same time. Race condition may have and there may be possibility that T_3 acquire the lock instead of T_2 . Then deadlock may happen, where T_3 waits indefinitely for T_2 to release the lock to M and T_2 waits indefinitely for T_3 to release the lock to N.



No. Kad Pengenalan/No. ISID:
I.C. No./ISID No.

No. Soalan.....
Question No.

Muka surat.....
Page

Jangan tulis apa-apa di ruangan ini
Do not write anything in this column

Transaction schedule when deadlock occurs

Time	T ₁	T ₂	T ₃
t ₁	begin transaction		
t ₂	lock-S(X)		
t ₃	read-item(X)		
t ₄	unlock(X)	begin transaction	
t ₅	lock-S(Y)	lock-X(M)	begin transaction
t ₆	read-item(Y)	read-item(M)	lock-X(N)
t ₇	unlock(Y)	request lock-S(N)	read-item(N)
t ₈	commit	WAIT	request lock-S(M)
t ₉		WAIT	WAIT
t ₁₀		WAIT	WAIT



No. Kad Pengenalan/No. ISID:.....
I.C. No./ISID No.

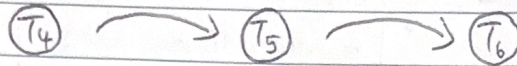
No. Soalan.....
Question No.

Muka surat.....
Page

Jangan tulis apa-apa di ruangan ini
Do not write anything in this column

Question 2

(i)



(ii) Since there is no cycle, the schedule is conflict serializability.

(iii)	Time	T ₄	T ₅	T ₆
	t ₁	begin transaction		
	t ₂	lock-X(P)		
	t ₃	read-item(P)		
	t ₄	lock-S(Q)		
	t ₅	read-item(Q)	begin transaction	
	t ₆	P = (P * 5) + Q	lock-X(Q)	
	t ₇	write-item(P)	WAIT	
	t ₈	commit/unlock(P, Q)	WAIT	
	t ₉		read-item(Q)	
	t ₁₀		lock-S(R)	
	t ₁₁		read-item(R)	begin transaction
	t ₁₂		Q = Q + R	lock-X(R)
	t ₁₃		write-item(Q)	WAIT
	t ₁₄		commit/unlock(R, Q)	WAIT
	t ₁₅			read-item(R)
	t ₁₆			R = R - 100
	t ₁₇			write-item(R)
	t ₁₈			commit/unlock(R)



No. Kad Pengenalan/No. ISID:.....
I.C. No./ISID No.

No. Soalan.....
Question No.

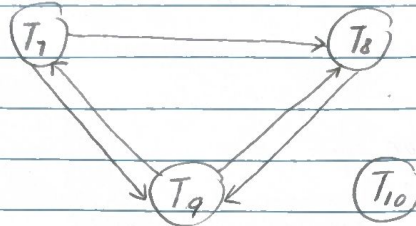
Muka surat.....
Page

Jangan tulis apa-apa di ruangan ini
Do not write anything in this column

Question 3

(a) (i) Non-serial schedule because multiple transactions execute concurrently and overlap in time.

(ii)



(iii) The schedule is not conflict-serialisable as there is cycles.

(iv) Lost Update Problem. T_9 overwrites the value of X written by T_7 ; T_8 overwrites the value of X written by T_9 .
Uncommitted Dependency. T_8 reads uncommit value of X written by T_7 that might be rolled back.

(b) (i)

T_{11}	T_{12}	T_{13}	T_{14}
begin transaction			
x-lock (A)			
read (A)	begin transaction		
	x-lock (B)		
	read (B)	begin transaction	
$A = A + 20$	$B = B - 50$	request x-lock(A)	begin transaction
write (A)			x-lock (C)
unlock (A)	write (B)		read (C)
	unlock (B)	x-lock (A)	$C = C + 10$
Commit / Rollback (if)		read (A)	write (C)
		$A = A - 30$	unlock (C)
	request x-lock(A)	write (A)	
		unlock (A)	
	x-lock (A)		
	read (A)		
	$A = A * 2$		
	write (A)	Commit	Commit
	unlock (A)		
	Commit		



No. Kad Pengenalan/No. ISID:.....
I.C. No./ISID No.

No. Soalan.....
Question No.

Muka surat.....4
Page

Jangan tulis apa-apa di ruangan ini
Do not write anything in this column

- (i) Uncommitted Dependency. T_{13} reads A while is last written by T_{11} and T_{11} is not committed yet. T_{11} may rollback and T_{13} will get the incorrect value of A . Besides, concurrency problems may still occur even though all transactions commit successfully

T_{11}	T_{12}	T_{13}	T_{14}
begin transaction			
write-lock (A)			
read (A)	begin transaction		
	write-lock (B)		
	read (B)	begin transaction	
$A = A + 200$	$B = B - 50$	write-lock (A)	begin transaction
		WAIT	write-lock (C)
write (A)		WAIT	read (C)
	write (B)	WAIT	$C = C + 10$
		WAIT	write (C)
commit/unlock (A)		WAIT	
	write-lock (A)	read (A)	
	WAIT	$A = A - 30$	
	WAIT	write (A)	
	WAIT	commit/unlock (A)	commit/unlock (C)
	read (A)		
	$A = A * 2$		
	write (A)		
	commit/unlock (A, B)		

- (iv) Limited Concurrency. The strict rules in 2PL can reduce the number of transactions that can run at the same time. This result in slow performance and long wait times.

No. Kad Pengenalan No. ISID:
 I.C. No. / ISID No.

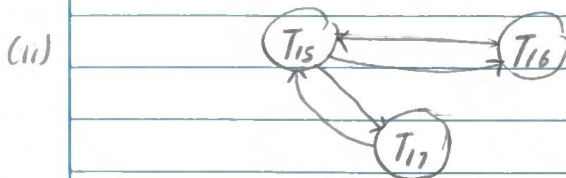
 No. Soalan
 Question No.

 Muka surat **5**
 Page

 Jangan tulis apa-apa di ruangan ini
 Do not write anything in this column

Question 4

- (i) Non-serial. The operation of multiple transactions are interleaved. read and write operation happens on the same item at the same time at t_4 and t_5 .



- (iii) The schedule is not conflict-serialisable as there are loops.
- (iv) Lost updates. At t_4 , T_{15} writes J and T_{17} reads J at the same time. At t_5 , T_{16} writes K and T_{15} reads K at the same time.

(v)

T_{15}	T_{16}	T_{17}
begin transaction		
write-lock (J)		
read (J)	begin transaction	
$J = J + 10$	write-lock (K)	
write (J)	read (K)	begin transaction
write-lock (K)	$K = K - 5$	write-lock (J)
WAIT	write (K)	WAIT
WAIT		WAIT
WAIT	commit / unlock (K)	WAIT
read (K)		WAIT
$K = K + 20$		WAIT
write (K)		WAIT
commit / unlock (J, K)		WAIT
		read (J)
		$J = J - 15$
		write (J)
		commit / unlock (J)